



## Defining features of linear relationships

1 Two variables that change together at a constant rate when plotted on a graph will form a linear relationship. Fill in the blank

2 Match the arithmetic sequences to the descriptors. Write the correct letter in each box

|   |                       |
|---|-----------------------|
| a | 11, 4, - 3, - 10, ... |
| b | 11, 7, 3, - 1, ...    |
| c | 7, 11, 15, 19, ...    |
| d | 7, 18, 29, 40, ...    |
| e | 11, 15, 19, 23, ...   |
| f | 7, 3, - 1, - 5, ...   |

|   |                                  |
|---|----------------------------------|
| d | Starts at 7 and increases by 11. |
| e | Starts at 11 and increases by 4. |
| f | Starts at 7 and decreases by 4.  |
| c | Starts at 7 and increases by 4.  |
| b | Starts at 11 and decreases by 4. |
| a | Starts at 11 and decreases by 7. |

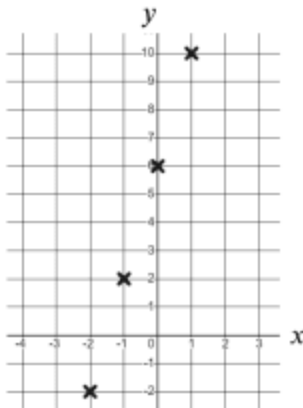
3 Match the common differences to the arithmetic sequences. Write the correct letter in each box

|   |                           |
|---|---------------------------|
| a | - 8, - 3, 2, 7, ...       |
| b | 7, 2, - 3, - 8, ...       |
| c | - 8, - 6, - 4, - 2, ...   |
| d | - 4, - 6, - 8, - 10, ...  |
| e | 4, 1, - 2, - 5, ...       |
| f | - 14, - 11, - 8, - 5, ... |

|   |     |
|---|-----|
| d | - 2 |
| e | - 3 |
| a | + 5 |
| c | + 2 |
| f | + 3 |
| b | - 5 |



4 Which of the below coordinates can you see plotted here? Tick **3** correct answers



☒  $(-2, -2)$

☐  $(-2, 2)$

☒  $(-1, 2)$

☐  $(6, 0)$

☒  $(0, 6)$

5 Here is an arithmetic (linear) sequence for which we are given the second and fourth terms. Which of the below are the missing terms? Tick **3** correct answers

\_\_, 0.6, \_\_, 0.78, \_\_, ...

☒ 0.87

☐ 0.5

☐ 0.8

☒ 0.69

☒ 0.51

6 For the linear relationship  $y = 15 - 7x$  what is the  $y$  coordinate when the  $x$  coordinate is 1.5? State your answer as a decimal. Fill in the blank

4.5

